

Machine-Learning Interatomic Potentials for Structural and Dynamical Properties of Ag₂₀ Clusters



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Introduction

Machine learning interatomic potentials (MLIPs) provide near-density functional theory (DFT) accuracy at a fraction of the computational cost. Ag₂₀ has attracted considerable attention because of its potential applications in catalysis, plasmonics, sensing, and nanophotonics, and its low-energy structures have been extensively investigated [1]. However, its complex potential energy surface, characterized by multiple low-energy isomers, makes accurate atomistic simulations computationally demanding. In this work, a SchNet model [2] implemented in the SchNetPack framework [3] was trained on DFT data to efficiently learn the potential energy surface of Ag₂₀, enabling fast and accurate structural and dynamical simulations.

Methods

Data generation

The Ag₂₀ dataset was constructed by combining normal-mode-displaced configurations generated around six low-energy isomers with graph-unique structures obtained from global structure searches, providing a diverse sampling of the configurational space [4].

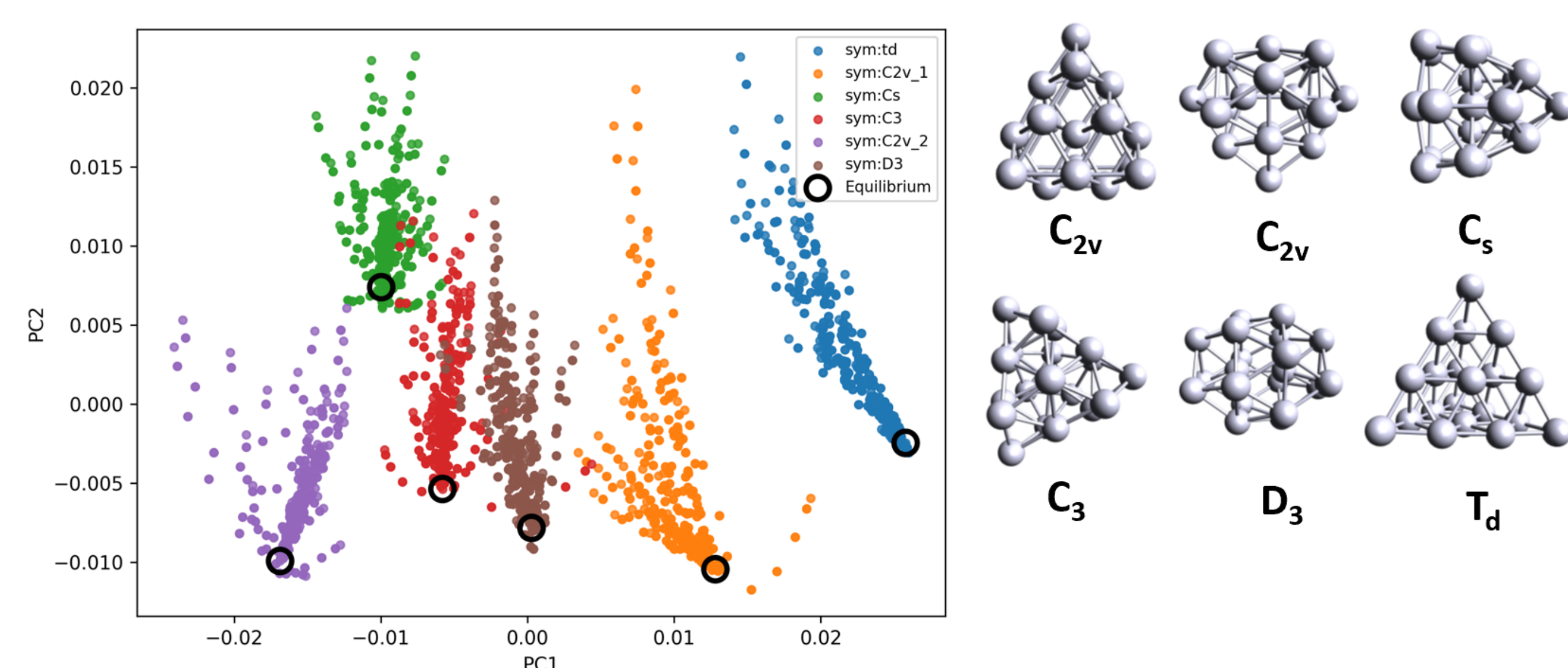


Figure 1: PCA projection of the configurational space sampled by normal-mode displacements around six low-energy Ag₂₀ isomers. Principal Component Analysis (PCA) projects high-dimensional atomic configurations into a two-dimensional space for visualization, while the Smooth Overlap of Atomic Positions (SOAP) descriptor provides the structural similarity metric used for Farthest Point Sampling (FPS) to select diverse configurations.

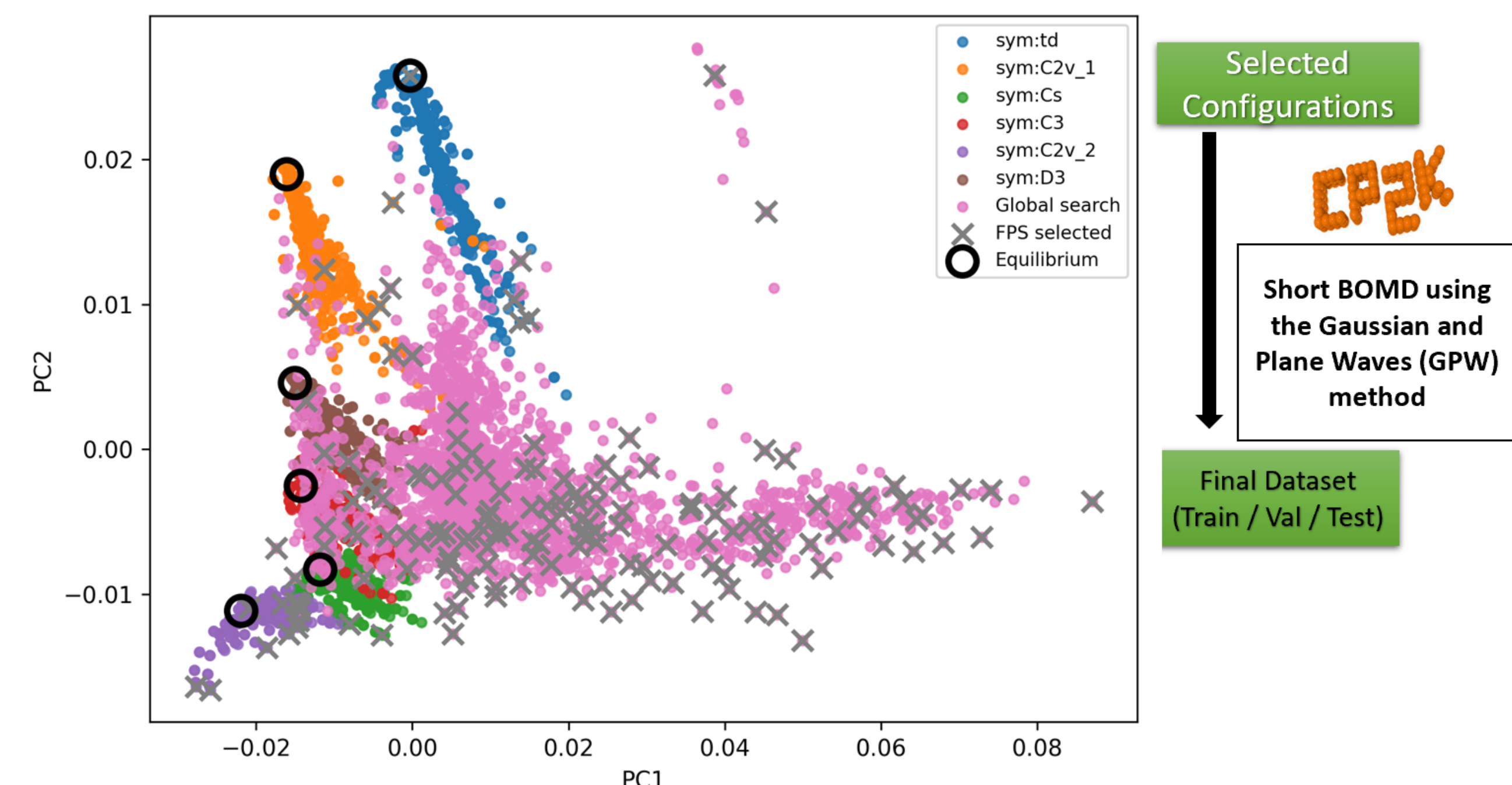


Figure 2: PCA projection of the complete Ag₂₀ dataset obtained from normal mode displaced configurations and graph-unique structures obtained from global structure searches. Farthest Point Sampling (FPS) was applied to select diverse configurations, each of which was propagated by short Born–Oppenheimer molecular dynamics (BOMD) simulations performed with the CP2K Quickstep module using the Gaussian and Plane Waves (GPW) formalism, the PBE exchange–correlation functional with Grimme’s D2 dispersion correction, and a DZVP basis set. Approximately 220 000 configurations were generated, of which about 30 000 were selected for the training, validation, and test sets.

Model performance

Property	Training set		Validation set		Test set	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
Energy (eV)	0.0228	0.0315	0.0232	0.0315	0.051	0.078
Atomic forces (eV/Å)	0.0112	0.0203	0.0135	0.0324	0.035	0.064
Dipole moment (Debye)	0.0051	0.0074	0.0048	0.0115	0.0132	0.0201

Table 1: MAE and RMSE of the trained models evaluated on the training, validation, and test sets.

Results

Relaxation

The relaxed geometry is obtained by iteratively updating the atomic coordinates,

$$\mathbf{R}^{(t+1)} = \mathbf{R}^{(t)} + \alpha \mathbf{F}^{(t)}, \quad \mathbf{F}^{(t)} = -\nabla_{\mathbf{R}} E_{\theta},$$

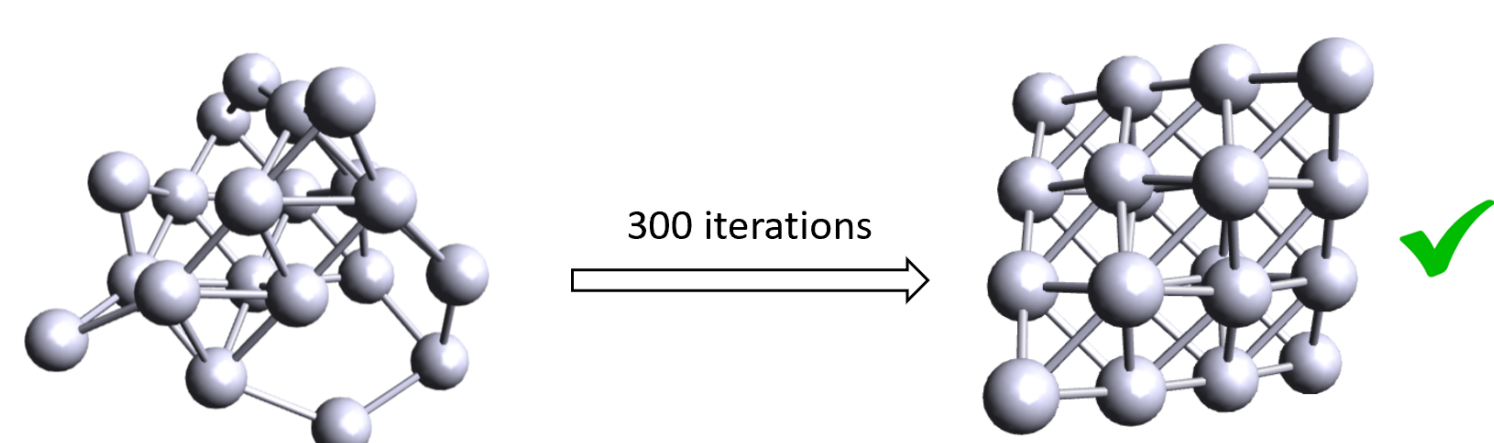


Figure 3: Relaxation of the perturbed T_d symmetry of Ag₂₀. The atomic coordinates $\mathbf{R}$ are iteratively updated using the forces $\mathbf{F}$ predicted by the trained SchNet potential, where α is the optimization step size and the relaxation is continued until the maximum atomic force falls below a predefined convergence threshold.

Local chemical potential

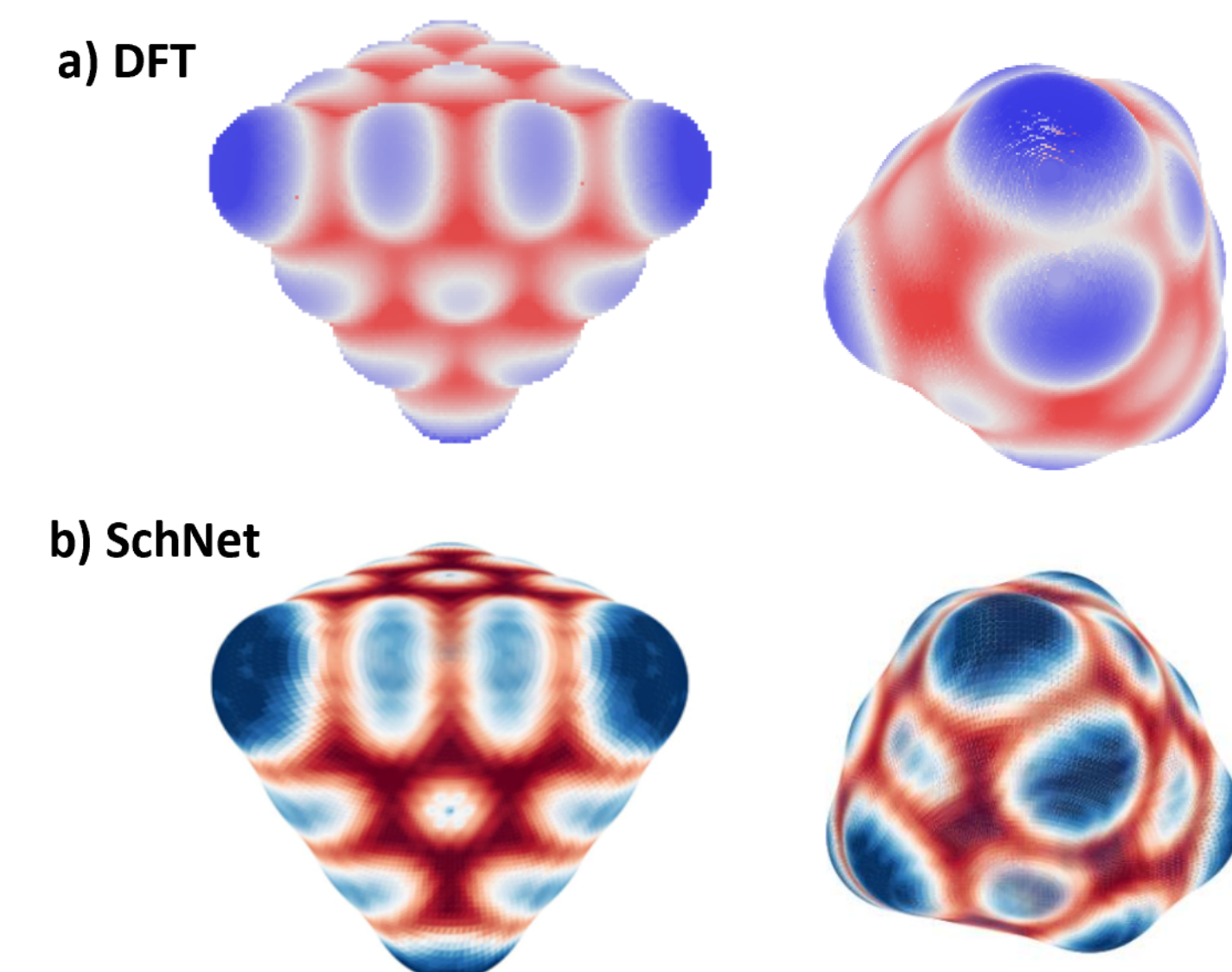


Figure 4: Comparison of the electrostatic potential (ESP) and the SchNet local chemical potential mapped onto the Ag₂₀ molecular isosurface. (a) DFT ESP calculated at the PBE-D3(BJ) level. (b) SchNet local chemical potential obtained using a virtual probing Ag atom.

Energy prediction: short- and long-range interactions

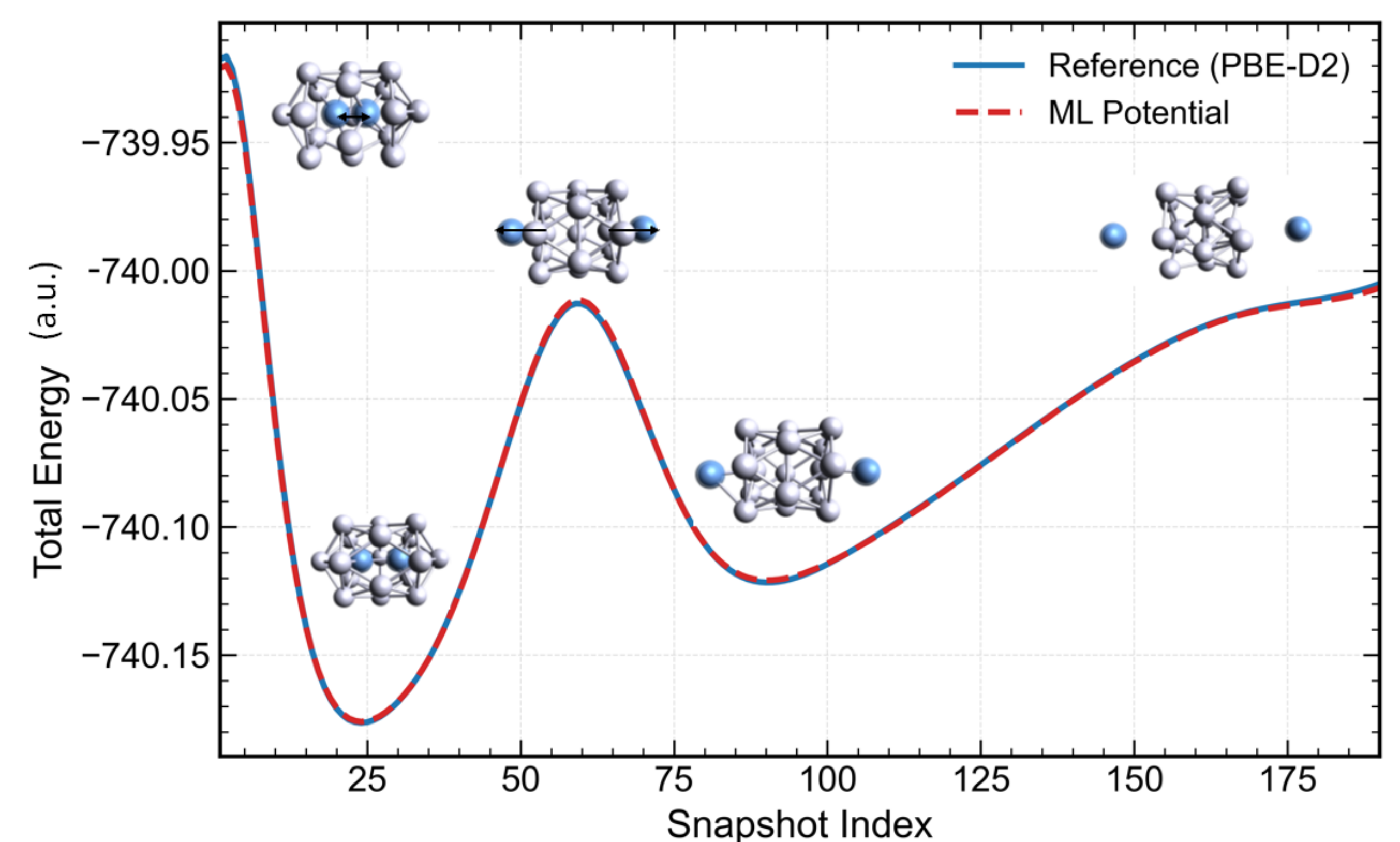


Figure 5: Comparison of the ML-predicted and DFT (PBE-D2) energies for representative Ag₂₀ configurations spanning short- and long-range interactions. The selected configurations illustrate the ability of the trained potential to accurately reproduce the DFT energy profile across the configurational pathway.

Molecular dynamics and IR spectroscopy

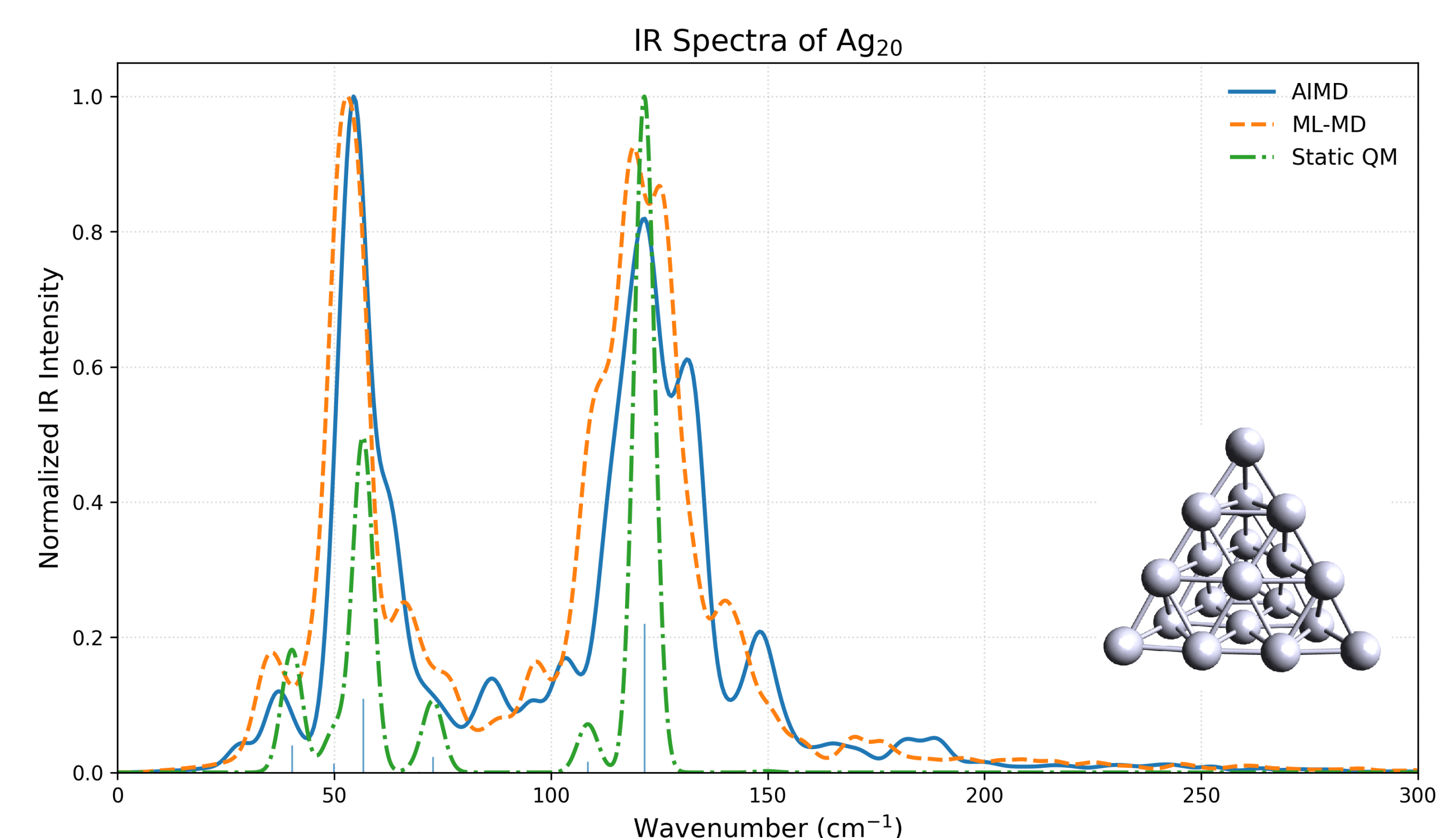


Figure 6: Comparison of the normalized IR spectra of the T_d Ag₂₀ isomer obtained from static QM calculations (BP86/def2-TZVP), PBE-D2 Born–Oppenheimer molecular dynamics (AIMD), and SchNet-based molecular dynamics (ML-MD). All spectra were independently normalized and Gaussian broadened using a full width at half maximum (FWHM) of 5.6 cm⁻¹.

References

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- [2] Kristof T. Schütt, Pieter-Jan Kindermans, Huziel E. Sauceda, Stefan Chmiela, Alexandre Tkatchenko, and Klaus-Robert Müller. SchNet: A continuous-filter convolutional neural network for modeling quantum interactions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
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